

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour

Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

*I V G ABBIRAMY - 192524097 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

V G ABBIRAMY

1. Water Jug Problem (4-Gallon and 3-Gallon Jugs)

Problem

Obtain exactly **2 gallons** of water in the **4-gallon jug** using only the pump and the two jugs.

Initial State

- 4-gallon jug = 0 gallons
- 3-gallon jug = 0 gallons

Goal State

- 4-gallon jug = **2 gallons**

Solution Steps

Step	Action	4-Gallon Jug	3-Gallon Jug
1	Fill the 3-gallon jug	0	3
2	Pour 3-gallon jug into 4-gallon jug	3	0
3	Fill the 3-gallon jug again	3	3
4	Pour water into 4-gallon jug until it is full (only 1 gallon needed)	4	2
5	Empty the 4-gallon jug	0	2
6	Pour the remaining 2 gallons from the 3-gallon jug into the 4-gallon jug	2	0

Answer

The **4-gallon jug** now contains exactly **2 gallons**, achieving the goal.

2. Mars Rover Agent

i. What are the percepts for this agent?

Percepts are the inputs received by the Mars Rover.

- Images from cameras
- Rock and soil composition
- Temperature
- Atmospheric pressure
- Terrain information
- Distance to obstacles
- Battery level
- Position and location

ii. Characterize the operating environment

- **Partially Observable** – Not all information is available.
- **Deterministic** – Actions usually produce predictable results.
- **Sequential** – Current actions affect future actions.
- **Dynamic** – Environment changes due to weather and terrain.
- **Continuous** – Movement and sensing occur continuously.
- **Single Agent** – Rover operates independently.

iii. What are the actions the agent can take?

- Move forward/backward
- Turn left/right
- Capture images
- Collect rock and soil samples
- Analyze samples
- Drill rocks

- Send data to Earth
- Avoid obstacles
- Recharge using solar panels

iv. How can one evaluate the performance of the agent?

Performance can be measured by:

- Number of successful samples collected
- Accuracy of scientific analysis
- Safe navigation
- Energy efficiency
- Distance covered
- Successful communication with Earth
- Mission completion rate

v. Suitable agent architecture and reason

Goal-Based Agent with Utility-Based features

Reason:

- Works towards scientific goals.
- Chooses actions that maximize mission success.
- Considers safety, energy usage, and time.
- Adapts to unexpected obstacles.

3. 8-Queens Problem

Problem Formulation

State Space

All possible arrangements of 8 queens on the chessboard.

Initial State

Empty 8×8 chessboard.

Actions

Place one queen in a safe position in the next column.

Transition Model

Move from one state to another by placing a queen without violating constraints.

Goal Test

A solution is reached when:

- All 8 queens are placed.
- No two queens attack each other horizontally, vertically, or diagonally.

Path Cost

Each move has equal cost (1).

Search Strategies

- **Breadth-First Search (BFS):** Explores all states level by level.
- **Depth-First Search (DFS):** Explores one branch deeply before backtracking.
- **Backtracking Search (Most Common):** Places queens one by one and backtracks when a conflict occurs.
- **Heuristic Search:** Uses heuristics to reduce unnecessary exploration.

4. OLA Cab Booking Problem

Type of Problem-Solving Agent

A **Goal-Based Agent** is suitable.

Reason:

- The customer's goal is to reach the destination.
- The agent evaluates available cab options (Mini, Micro, Sedan, Prime, Shared, etc.) and selects one that satisfies the user's preferences such as comfort, cost, and availability.

Pseudocode

Algorithm OLA_Cab_Booking

Start

Input source location

Input destination location

Display available cab types

(Micro, Mini, Sedan, Prime, Shared)

Customer selects preferred cab type

If selected cab is available then

 Calculate fare

 Display fare and estimated arrival time

 Confirm booking

 Assign nearest driver

 Start trip

Else

 Display "Selected cab not available"

 Suggest another available cab

End If

Complete trip

Generate bill

Collect payment

Stop

Conclusion

A **Goal-Based Agent** is appropriate because it selects the most suitable cab based on the customer's objective of reaching the destination while considering available options and constraints.

5. Uniform Cost Search (UCS) – Least Cost Path

Given Weighted Graph

From the figure:

- $S \rightarrow A = 1$
- $S \rightarrow G = 12$
- $A \rightarrow B = 3$
- $A \rightarrow C = 1$
- $B \rightarrow D = 3$
- $C \rightarrow D = 1$
- $C \rightarrow G = 2$
- $D \rightarrow G = 3$

Uniform Cost Search (UCS) Steps

Step	Expanded Node	Path Cost	Frontier
1	S	0	A(1), G(12)
2	A	1	C(2), B(4), G(12)
3	C	2	D(3), G(4), B(4)

Step	Expanded Node	Path Cost	Frontier
4	D	3	G(4), B(4), G(6)
5	G	4	Goal Reached

Least-Cost Path

S → A → C → G

Total Cost

$$1 + 1 + 2 = 4$$

Search Tree

S(0)



Final Answer

- **Algorithm Used:** Uniform Cost Search (UCS)
- **Least-Cost Path:** **S → A → C → G**
- **Minimum Cost:** **4**

Conclusion:

UCS expands nodes in order of increasing path cost. It finds the optimal (least-cost) route from **S** to **G**, which is **S → A → C → G** with a total cost of **4**.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem